Nicolas Bailliet

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Summary

Third-year Computer Science student at ENSEEIHT, with a background in mathematics and numerical optimization. I train language models from scratch, on a single desktop GPU. My latest, HilbertLM, was written end to end: custom tokenizer, pre-training, fine-tuning, then a live chatbot anyone can talk to. Now I want to build and deploy LLM and agentic systems in production, and fine-tune models for the teams that use them.

Seeking a 6-month internship outside France from mid-March 2027 · Based in Paris · Open to relocation · EU citizen

Experience

Research Engineer Intern — Optimal Control & Scientific Computing

IRIT / ENSEEIHT, Toulouse | Jun 2025 – Aug 2025

Technologies: Julia, JuMP

- Enhanced and validated an open-source Julia optimal-control solver library, across two axes: software maintenance (bug fixing) and numerical validation.
- Built an automated benchmark looping over 15 control problems: convergence checks for both the solver and JuMP (reference Julia modeling language), initialization comparison, and relative error in the L^2 norm on the state x and control u between the two solvers.
- Contributed unit tests & API documentation via GitHub (commits, Pull Requests, peer reviews), positive team feedback for surfacing critical bugs.

Education

Computer Science & Engineering — ENSEEIHT (INP Toulouse)

Sep 2023 – Present

ML algorithms, numerical methods, optimization, HPC, big data systems.

Preparatory School, Maths & Physics — Lycée Chaptal, Paris

Sep 2020 – Jul 2023

Linear algebra, real analysis, probability theory, differential equations.

Projects

HilbertLM (135M & 163M)

Live demo: baillietn.github.io/HilbertLM-Lab/ | HuggingFace: huggingface.co/collections/baillietn/hilbertlm

135M & 163M params | 20B tokens trained | ~70h · RTX 5080 | 79K tok/s throughput | 4 models on HF Hub

Technologies: Python, PyTorch, Transformer Engine, Flash Attention 3, FP8, HuggingFace, BPE

Two decoder-only transformers built entirely from scratch. 30-layer architecture: GQA (9Q/3KV heads), RoPE, SwiGLU — custom BPE tokenizer (49K vocab). 4-stage curriculum pre-training (FineWeb-Edu, Python-Edu, FineMath, Cosmopedia-v2) + SFT on 100M conversational tokens (SmolTalk). Loss landscape visualization via PCA on checkpoints. Deployed as Base & Chat variants (BF16 + FP8) on HuggingFace Hub with streaming chatbot interface.

Skills

Machine Learning & LLMs: LLM architecture (GQA · RoPE · SwiGLU), Pre-training pipeline (data · curriculum · BPE), Fine-tuning (SFT), Mixed-precision training (BF16/FP8), Inference & deployment (HF)

Frameworks & Tools: Python, PyTorch, HuggingFace, Transformer Engine, Flash Attention, FP8 / BF16, JAX/Flax, NumPy, scikit-learn, Julia, SQL, Git, Linux, HPC

Languages

French: Native, English: Professional